



COMPUTER NETWORKS

Computer Networks and the Internet

Team Networks

Department of Computer Science and Engineering

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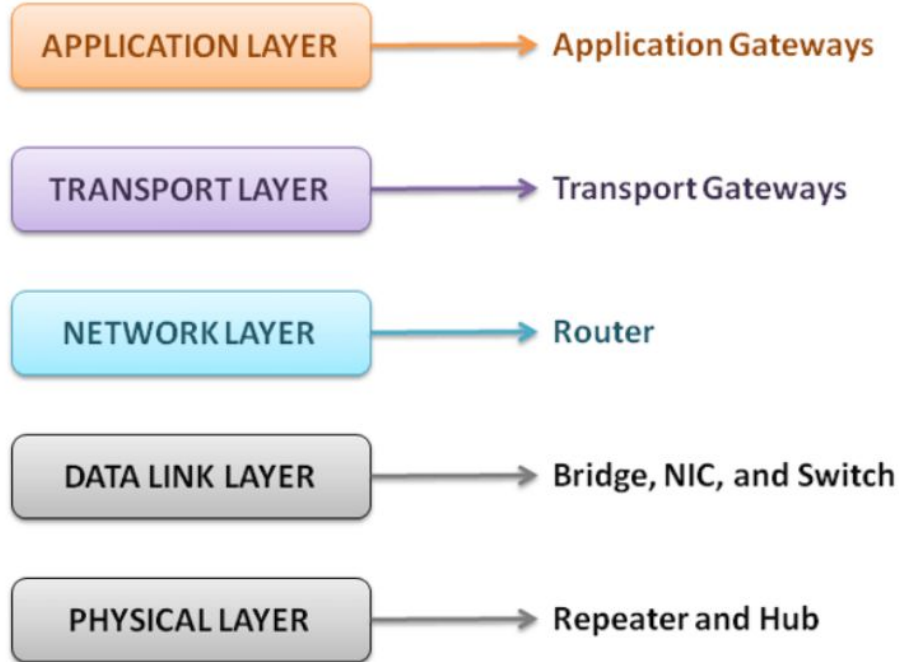
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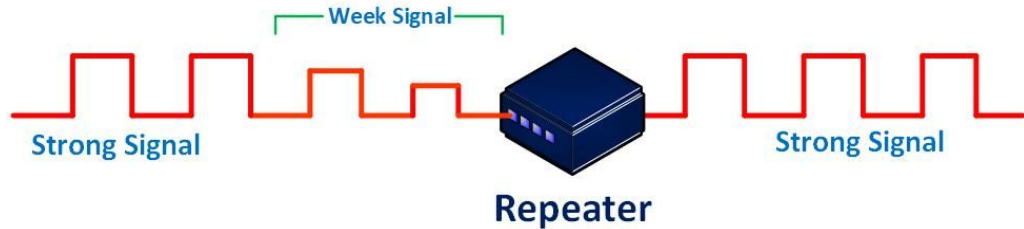
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Introduction to Network Devices



Repeater

- Regenerate the signal over the same network before the signal becomes too weak or corrupted to extend the length to which the signal can be transmitted over the same network
- Works at the Physical Layer



Working of a Repeater:-

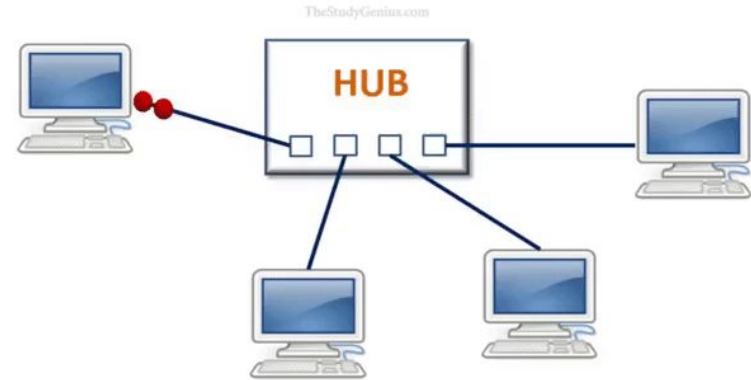
1. Signal Reception: Captures incoming data signal (electrical voltages or electromagnetic waves).
2. Signal Amplification: Strengthens weakened signals due to attenuation and interference.
3. Signal Regeneration: Reconstructs the signal to ensure data integrity and network compliance.
4. Signal Transmission: Sends the amplified and regenerated signal to its target, minimizing further deterioration.

Hub

- Basic networking device that is used to connect multiple devices together on a local network.
- It shares bandwidth among all connected devices and broadcasts data sent by one device to all others, giving each device equal access to the network
- Operates at Physical Layer

Features of a Hub:-

1. Always Broadcast: Hubs broadcast data to all connected devices, regardless of the intended recipient.
2. Half-duplex Communication: Only one device can transmit at once.
3. Cannot Store MAC Addresses: Hubs lack intelligence and cannot store MAC addresses or maintain tables.
4. No Data Filtering: Hubs transmit all data to all connected devices without filtering.



Modem

- Device that connects a computer to the internet by converting digital signals to analog and vice versa for data transmission.
- Operates at the Physical Layer

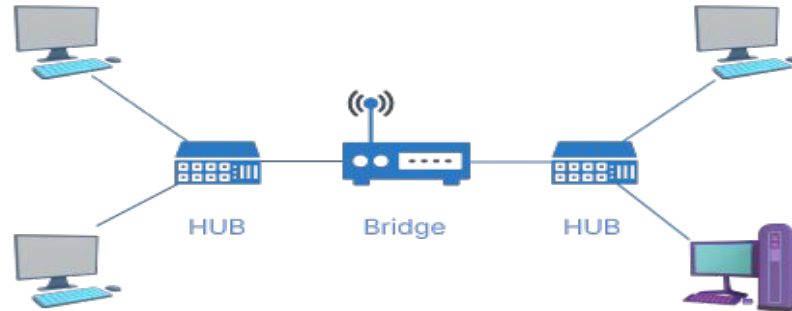


Modem Data Transmission Process:-

1. Modulation: The modem converts digital data from the computer into an analog signal for transmission.
2. Transmission: The modulated signal is sent over the communication line to the receiving modem.
3. Demodulation: The receiving modem converts the analog signal back into digital data.
4. Decoding: The digital data is then sent to the computer for use.

Bridge

- A 2-port device that joins two or more network segments and routes data between them according to MAC addresses
- Operates at the Data Link Layer



Functions :-

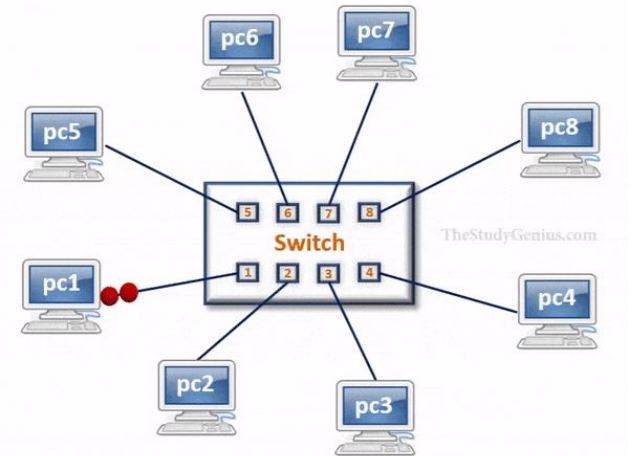
1. Segmentation: Divides a large network into smaller segments to reduce congestion and improve performance.
2. Filtering: Uses MAC addresses to selectively forward packets, enhancing security and bandwidth usage.
3. Learning: Tracks MAC addresses and port mappings to optimize packet forwarding.
4. Forwarding: Directs packets between segments, reducing unnecessary traffic and ensuring proper delivery.

Switch

- An intelligent network device, functioning as a multiport bridge, uses MAC addresses to forward data packets to the appropriate destination ports.
- Operates at Data Link Layer

Features of a Switch:-

1. Employs packet switching to receive and route data from source to destination.
2. Can perform some error checking before forwarding data to the destined port.
3. Can operate in full-duplex mode, enabling simultaneous data transmission & reception in network
4. Store MAC addresses of connected devices, routing communication only to the intended destination, reducing collisions and eliminating broadcast domains.



NIC - Network Interface Card

- It is a physical card or chip, which contains MAC addresses, helps to identify the device on the network.
- Used at the Data Link Layer

Functions of a NIC:-

1. Connectivity: Links the device to the network (wired or wireless).
2. Data Transmission: Sends and receives data between devices.
3. Data Conversion: Converts data for transmission and reception.
4. Error Handling: Detects and corrects transmission errors.
5. Traffic Management: Prioritizes or limits data to avoid congestion.
6. Security: Provides encryption and authentication for protection.

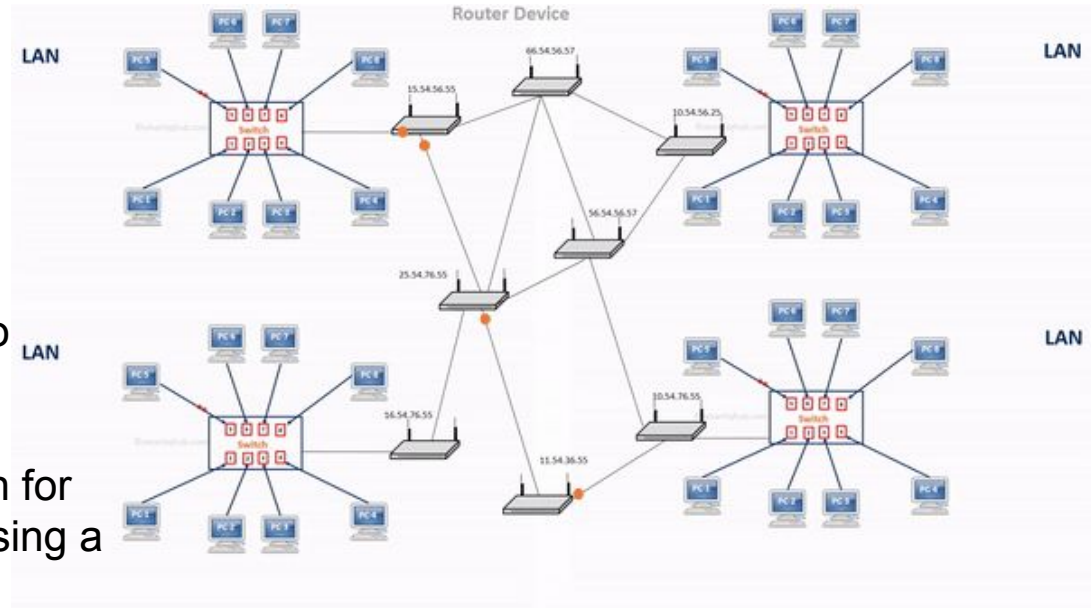


Router

- Device that routes/forwards data packets based on their IP addresses and have a dynamically updating routing table based on which they make decisions on routing the incoming packets.
- Operates at the Network Layer

Functions of a Router:-

1. Forwarding: Receives packets, checks headers, performs basic functions, and forwards packets to the appropriate output port.
2. Routing: Determines the best path for a packet to reach its destination using a routing table built with algorithms.



Router

Functions of a Router:-

3. Network Address Translation (NAT): Translates between private and public IP addresses, enabling devices on a private network to access the internet.

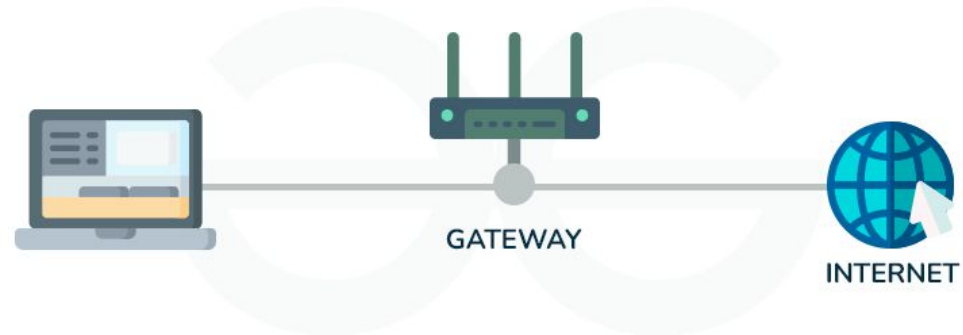


Gateway

- Network connectivity device that connects two different configuration networks
- The gateway acts as a portal between two applications via protocol communications, allowing them to share data on the same or other systems. Hence, a gateway is also a protocol converter that can operate at any OSI model layer.

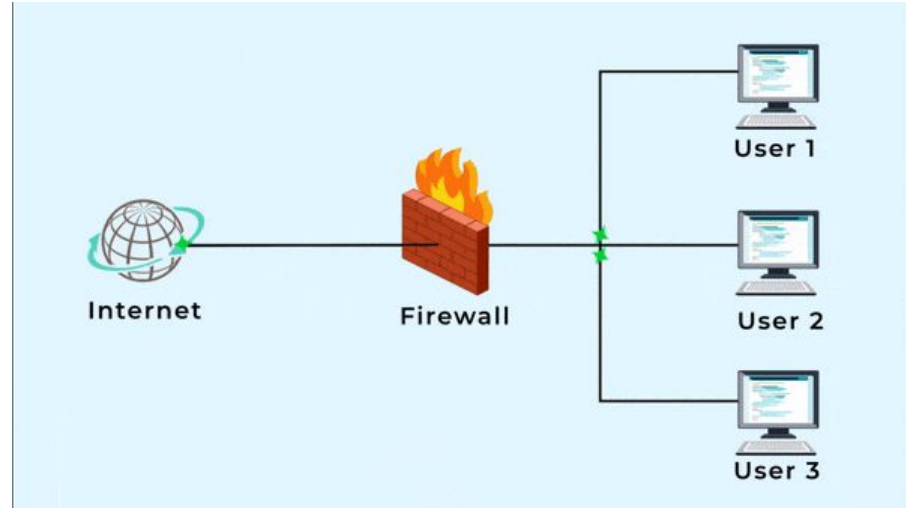
Types of Gateway:-

1. Unidirectional Gateway: Data flows in one direction, with changes replicated from source to destination only.
2. Bidirectional Gateway: Data flows in both directions, used for synchronization.



Firewall

- Network security device that monitors incoming and outgoing network traffic and decides whether to allow or block specific traffic based on a defined set of security rules
- It can be used in the transport layer or the application layer



Feature	Hub	Switch
Function	Broadcasts data to all devices	Forwards data to the specific destination device based on MAC address
Layer	Operates at Physical Layer	Operates at Data Link Layer
Data Handling	Sends data to all connected devices	Sends data only to the intended device
Collision Domain	Single collision domain	Multiple collision domains (one per port)
Bandwidth	Shared bandwidth among all devices	Dedicated bandwidth for each device
Intelligence	Basic, no intelligence	More intelligent, stores MAC addresses and makes forwarding decisions
Data Filtering	No filtering, all data is broadcasted	Filters data based on MAC addresses
Usage	Suitable for small networks	Suitable for larger, more complex networks
Performance	Lower performance due to collisions	Higher performance with reduced collisions

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Introduction to Network Devices

Feature	Switch	Router
Definition	A network device that connects multiple devices within the same network.	A network device that connects multiple networks and routes data between them.
Data Transmission	Uses MAC addresses to forward data within the network.	Uses IP addresses to route data between networks.
Function	Operates at Data Link Layer	Operates at Network Layer
Broadcast Domain	Typically operates within a single broadcast domain.	Breaks up broadcast domains and creates separate ones.
Usage	Used in LANs to expand network capacity.	Used to connect LANs to the internet or other LANs.
Speed	High-speed data transfer within the network.	Typically slower due to complex routing processes.



THANK YOU

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